

Jacobian

If 6 joints and 6 DOF

$$\begin{matrix} \text{Trans} \\ \text{Rot} \end{matrix} \left\{ \begin{matrix} T_0 dx \\ T_0 dy \\ T_0 dz \\ T_0 \partial x \\ T_0 \partial y \\ T_0 \partial z \end{matrix} \right\} = \begin{bmatrix} T_0 J_{11} & T_0 J_{12} & & & & \\ T_0 J_{21} & & \ddots & & & \\ & & & \ddots & & \\ & & & & \ddots & \\ & & & & & \ddots \\ & & & & & T_0 J_{66} \end{bmatrix} \begin{bmatrix} d\theta_1 \\ d\theta_2 \\ d\theta_3 \\ d\theta_4 \\ d\theta_5 \\ d\theta_6 \end{bmatrix} \begin{matrix} \rightarrow \text{1st joint} \\ \\ \\ \\ \rightarrow \text{6th joint} \end{matrix}$$

$$T_0 dx = J_0 J_{11} d\theta_1 + J_0 J_{12} d\theta_2 + \dots + J_0 J_{16} d\theta_6$$

If Joint 1 is Trans:

$$J_1 = \begin{bmatrix} n_z \\ o_z \\ a_z \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ which means } dx = d\theta_1 \hat{z} \text{ (map to } z\text{-axis)}$$

n_z, o_z, a_z from $A_1 A_2 A_3 A_4 A_5 A_6$

If Joint 1 is Rot:

$$J_1 = \begin{bmatrix} \hat{z} \times p \\ n_z \\ o_z \\ a_z \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ p_x & p_y & p_z \end{bmatrix}$$

$$= -p_y \hat{i} + p_x \hat{j}$$

$$= \begin{bmatrix} -p_y \\ p_x \\ 0 \end{bmatrix}$$

$$\Rightarrow J_1 = \begin{bmatrix} -p_y \\ p_x \\ 0 \\ n_z \\ 0_z \\ a_z \end{bmatrix}$$